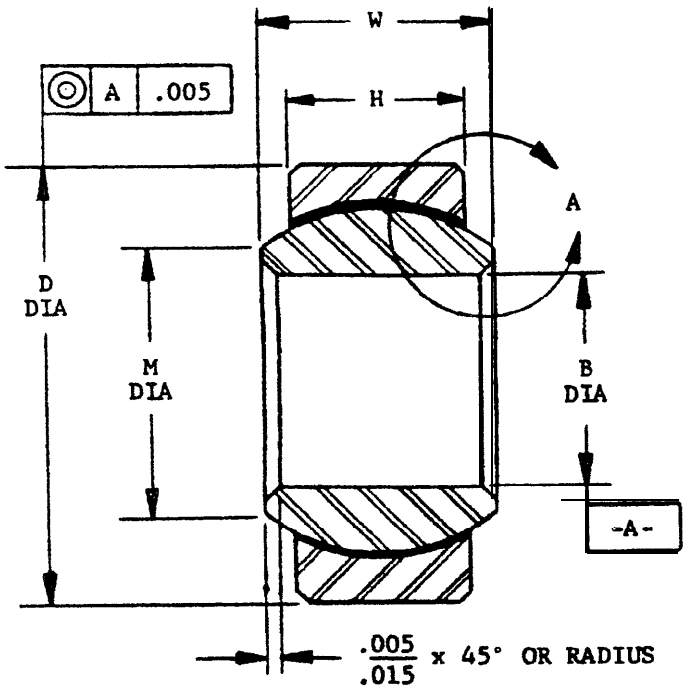
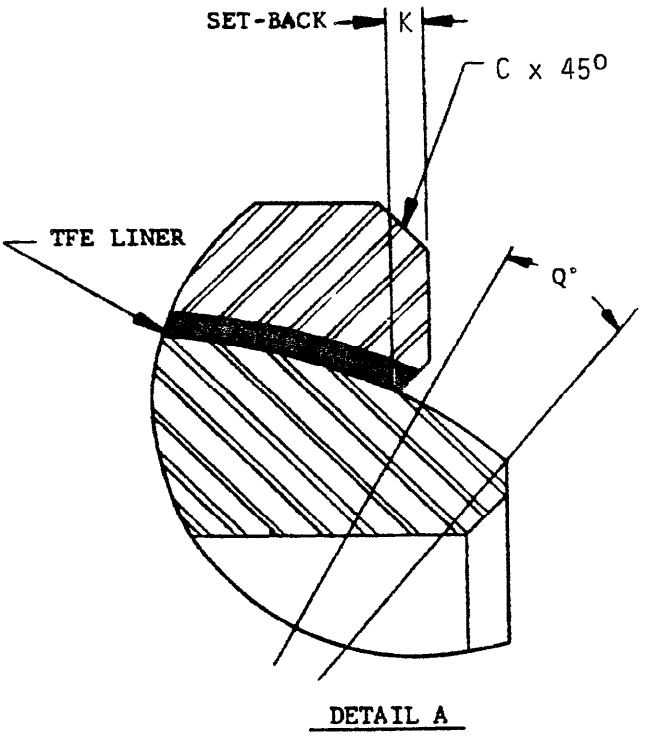
MILITARY SPECIFICATION SHEET

BEARINGS, PLAIN, SLIDING SPHERICAL, SELF-LUBRICATING, SELF-ALIGNING, HIGH SPEED OSCILLATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the bearings described herein shall consist of this document and the latest issue of MIL-B-81819.





DETAIL A

FSC 3120

Source: https://assist.dla.mil -- Downloaded: 202610-23T06:13Z
Check the source to verify that this is the current version before use.

TEST CONDITION NUMBER DASH NO.	K MAX	C ± .005	B + .0000 - .0005	D + .0000 - .0005	H ± .0005	M MIN	Q MIN	W + .000 - .002	NO-LOAD ROTATIONAL BREAKAWAY TORQUE (IN. LB)	WT LB (MAX)	QUALIFICATION TESTS									
											STATIC LIMIT LOAD		CONSTANT LOAD OSCILLATION WEAR TEST CONDITIONS				CYCLIC LOAD OSCILLATION WEAR TEST CONDITIONS 5/			
											RADIAL (LB) 1/	AXIAL (LB) 2/	TEST LOAD (LBS) 3/	TEST SPEED (CPM) 4/	OSC. ANGLE (DEGREES)	TEST TIME (HOURS)	TEST LOAD (LBS) 6/	TEST SPEED (CPM)	OSC. ANGLE (DEGREES)	TEST TIME (HOURS)
4	.035	.025	.3750	.8125	.312	.475	9	.406	1-15	.038	10800	1200	340	1500	± 6	800				
3	.040	.025	.6250	1.1875	.500	.739	8	.625	1-15	.110	29450	3600	900	1000	± 6	800				
2	.045	.035	1.0000	1.7500	.797	1.118	9	1.000	1-24	.386	75100	10200	2300	300	± 10	600	±2300 2300 ± 2300	300	±10	600
1	.045	.045	2.0000	3.2500	1.375	2.211	13	1.937	1-40	3.00	256600	33700	7800	200	± 7	750				

1/ Radial static limit load (pounds) = 68000 dh
Where d = nominal ball diameter
h = minimum liner width

2/ Axial static limit load (pounds) = $\frac{26000\ rt.^2}{4}$

3/ Load for constant oscillation wear test shall be applied with load link in compression using housing design specified in MIL-B-81819.

4/ One cycle shall consist of rotation from zero to +Ø degrees, return to zero to -Ø degrees and return to zero.

5/ Cyclic load oscillation wear tests shall be conducted on M81819/1-2 bearings. Satisfactory performance of -2 bearing in the cyclic load tests is adequate evidence that sizes -1, -3 and -4 will meet the requirements of cyclic load tests.

6/ Cyclic load tests shall be conducted so that the maximum load and maximum ball surface speed occur at the same time. Both Type I and Type II cyclic load tests are required.

TEST TYPE	RELATIONSHIP BETWEEN CYCLIC LOAD AND OSCILLATION ANGLE DURING ONE CYCLE OF OPERATION.
I	0 lbs → +2300 lbs → 0 lbs → −2300 lbs → 0 lbs $+10^{o} \rightarrow 0^{o} \rightarrow -10^{o} \rightarrow 0^{o} \rightarrow +10^{o}$
II	2300 lbs → 4600 lbs → 2300 lbs → 0 lbs → 2300 lbs $+10^{o} \rightarrow 0^{o} \rightarrow -10^{o} \rightarrow 0^{o} \rightarrow +10^{o}$

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:13Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

1. Material

Ball - 440C in accordance with AMS 5630 (plating or coating optional:
Outer Ring - 17-4PH in accordance with AMS 5643

2. Finish

Surface finish ball dia. - Ra8 Max.
Bore, Ball face and outer race dia. - Ra 32 Max.
All others - Ra 125 max.

3. Hardness

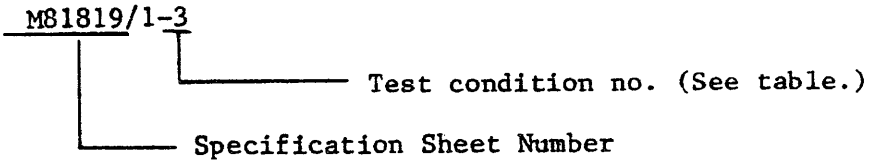
Ball - RC 55-62
Outer ring - RC 28 min., RC 37 max. before swaging

4. Passivation optional

5. Dimensions are in inches.

6. Break sharp edges and corners. Remove all burrs and slivers.

Example of part number



Custodians:
Navy - AS
Air Force - 11
Army - AV

Review:
DLA - IS

Preparing Activity:
Navy - AS
(project No. 3120-0605)

*US. GOVERNMENT PRINTING OFFICE: 1983-605-034/900

Source: https://assist.dla.mil -- Downloaded: 202610-23T06:13Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER (Number) AND TITLE
MIL-B-81819/1 Bearings, Plain, Slid. Spher., Self-Lub.

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426

Replaces edition of 1 Jan 72 which may be used.

S/N 0102-LF-001-4260

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:13Z
Check the source to verify that this is the current version before use.

)LD

EPARTMENT OF THE NAVY

OFFICIAL BUSINESS
:NALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Engineering Specifications and Standards
Department
Naval Air Engineering Center
Lakehurst, NJ 08733

)LD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:13Z
Check the source to verify that this is the current version before use.